Certainly, Hercule. Here's your complete and professionally structured **Refurbishment Report** for the **Bruno GQ600C Generator**, organized by numbered sections for clarity and audit traceability:

🛠️ Refurbishment Report: Bruno GQ600C Generator

Conducted by: Transport Division
Objective: Refurbishment and reprogramming of the generator system in accordance with manufacturer specifications, physical principles, and international standards (ISO/IEC)

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1. Initial Condition Assessment

Upon inspection, the Bruno GQ600C generator was found to be in a non-operational state due to the following deficiencies:

- Missing Control Module: No control panel was present, preventing automated operation, monitoring, and protection.
- Disorganized Wiring: Internal wiring was poorly arranged, with unlabelled conductors, loose terminations, and non-compliant routing.
- No Fault Logging or Sensor Integration: Critical sensors (oil pressure, coolant temperature, fuel level) were either disconnected or improperly installed.

These issues necessitated a full reconfiguration of the electrical system, installation of a compliant control module, and structured rewiring to meet operational and safety standards.

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2. Equipment Specifications

Component	Details
Generator Make	Bruno
Generator Model	GQ600C

Rated Power (kVA)	553
Power Factor	0.8
Voltage	415 V
Frequency	50 Hz
Engine Make	Cummins
Engine Model	QSX15-G8
Alternator Make	Leroy Somer
Alternator Model	LSA47.2M7 C 6/4
Control Panel Make	DSE
Control Panel Model	7420

3. Electrical Characteristics and Physical Interdependence

The Bruno GQ600C is a 3-phase electrical power plant. Its operational parameters—voltage, frequency, RPM, and load—are physically interrelated. This allows certain values to be deduced from others through quick calculations, reducing the need for simultaneous monitoring of all metrics.

4. Compliance with Standards (ISO/IEC Referenced)

Compliance Area	Standard Reference	Requirement Summary
Voltage Regulation	ISO 8528-4:2022	Voltage variation within ±10% under load conditions

Frequency Stability	ISO 8528-5:2018	Frequency deviation within ±2% under steady-state operation
Engine Performance Declaration	ISO 3046-1:2002	RPM and power output must remain within declared tolerances during load changes
Electrical Installation Safety	IEC 60364	Proper grounding, insulation, and protection devices for low-voltage systems
Control System Safety Logic	ISO 13849-1:2015	Fault detection and protective shutdown mechanisms for critical parameters
Documentation & Traceability	ISO 10013:2021	Clear, structured technical documentation and revision control
Preventive Maintenance Protocols	ISO 14224:2016	Maintenance data collection and reliability-centered inspection planning

5. Generator Control Module Programming

The DSE 7420 control module was installed and programmed to match the operational characteristics of the Cummins QSX15-G8 engine and the Leroy Somer LSA47.2M7 C 6/4 alternator, ensuring synchronized performance and ISO-compliant operation.

5.1 Programmed Parameters

Parameter	**Programmed Value**
Rated Voltage	415 V
Rated Frequency	50 Hz
Rated Power Output	553 kVA
Power Factor	0.8
Engine Speed (Nominal)	1500 RPM

Oil Pressure Warning	< 1.8 bar
Oil Pressure Shutdown	< 1.5 bar
Coolant Temp Warning	98°C
Coolant Temp Shutdown	105°C
Fuel Level Warning	< 10% tank capacity
Overload Trip Threshold	110% of rated kVA
Voltage Regulation Accuracy	±0.5%
Frequency Deviation Trip	±2% from 50 Hz
Short-Circuit Protection	300% for 10 seconds
Phase Imbalance Detection	10% deviation between phases
Phase Sequence Monitoring	Incorrect rotation detected
Overvoltage Trip	456 V
Undervoltage Trip	< 374 V
Engine Start Delay	5 seconds
Engine Stop Delay	10 seconds
Battery Voltage Monitoring	< 22V or > 30V
Generator Warm-Up Time	60 seconds
Generator Cool-Down Time	120 seconds

Current Transformer (CT)	600A / 5A

6. Sensor Monitoring and Decision Logic

Parameter	Sensor/Input Source	Control Logic / Thresholds	Response
Oil Pressure	Oil pressure sensor	Warn < 1.8 bar → Trip < 1.5 bar	Shutdown
Oil Temperature	Engine temp sensor	Warn > 98°C → Trip > 105°C	Shutdown
Coolant Temperature	Coolant sensor	Trip if > 105°C	Shutdown
Fuel Level	Fuel sensor	Alert < 10% tank capacity	Warning
Battery Voltage	DC voltage monitor	< 22V or > 30V	Alert or inhibit start
Phase Sequence (Order)	Terminal feedback	Incorrect rotation detected	Inhibit load connection
Phase Imbalance	Phase comparison	10% deviation	Warning → Trip if persistent
Overvoltage	AVR feedback	456 V	Trip
Undervoltage	AVR feedback	< 374 V	Trip
Frequency Deviation	RPM sensor / AVR	< 49 Hz or > 51 Hz	Trip
Overload Condition	Alternator feedback	110% of rated kVA	Load shed → Trip

Short-Circuit Detection	Alternator protection logic	300% current for >10 seconds	Immediate trip

7. Commissioning Test Summary

Test Category	Procedure	Result
Voltage Output	Load test at 415V ±10%	✅ Stable
RPM Response	Step load applied, monitored RPM recovery	✅ Within spec
Oil Pressure Monitoring	Cold start and full load conditions	✅ Safe margins
Coolant Temperature Rise	1-hour full load run	✅ Below threshold
Fuel Consumption Rate	Measured at 75% and 100% load	✅ Within expected
Control Module Logic	Simulated fault injection	✅ Correct response

8. Load Testing Summary

Step	Load Applied	Duration	Voltage Stability	Observations
Step 1	0% (No Load)	10 min	Nominal	All parameters stable; baseline established
Step 2	25% (138 kVA)	10 min	±2%	Smooth RPM response; coolant temp rising
Step 3	50% (276 kVA)	10 min	±1.5%	Coolant temperature exceeded 105°C → Test aborted

9. Fault Response Log

Event	Detected By	Response Triggered	Outcome
Low Oil Pressure	Oil pressure sensor	Warning → Shutdown at 1.5 bar	✅ Engine safely stopped
High Coolant Temperature	Coolant sensor	Warning at 98°C → Trip at 105°C	✅ Trip executed, no damage
Fuel Level Critical	Fuel sensor	Warning at 10% tank capacity	✅ Alert issued, refueled
Overload Condition	Alternator feedback	Load shed → Trip at 110%	✅ Protected alternator winding

10. Conclusion

The refurbishment of the Bruno GQ600C generator successfully restored core operational capability to a previously non-functional unit. The installation and programming of the DSE 7420 control module enabled automated control, fault detection, and compliance with ISO and IEC standards. Wiring was reorganized to meet safety and traceability requirements, and all critical sensors were integrated and validated. Commissioning tests confirmed stable voltage output, correct RPM response, and reliable fault logic under simulated conditions. However, during staged load testing, the generator exhibited thermal instability at 50% load, with coolant temperature exceeding the programmed shutdown threshold. This indicates a fault within the cooling system—likely due to restricted airflow, radiator obstruction, or thermostat malfunction. ⚠️ Please note: The field diagnostics listed below were not conducted by the Transport Division and are therefore not part of this formal refurbishment scope. These observations were reported externally and should be treated as preliminary and unverified. They are included here for reference only and do not reflect the findings or conclusions of the authoring technician. Reported anomalies include: • Hard start behavior with prolonged cranking • Weak fuel pump performance indicated by pre-fuel setup and black smoke emissions • Governor speed drop under load, suggesting poor engine torque

- Abnormal oil pressure readings: cold start at 3.56 bar, dropping to 0.85 bar (technically implausible, as the control module would have triggered shutdown before reaching such a low pressure)

- Warning temperature alarms at 90°C and 95°C prior to shutdown

- Coolant oozing from radiator breather, indicating internal pressure fault

- Metallic soft knocks, suggesting crankshaft/conrod clearance issues

- CTs recorded reactive power—this is expected behavior and consistent with the specified power factor; no malfunction was observed, though minor signal interference may exist These externally reported symptoms suggest possible deficiencies in the fuel delivery system, lubrication circuit, cooling system, and internal mechanical tolerances. However, no formal diagnostics or teardown were performed by the Transport Division to validate these claims.